Stochastic Dual Dynamic Programming vs. Actor-Critic Reinforcement Learning for Hydrothermal Scheduling: A Comparative Study on the Brazilian Interconnected Power System

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1 Introduction

The hydrothermal generation scheduling problem is one of the large-scale multistage stochastic optimization problems in power systems operations. Brazil's National Interconnected System (SIN) is particularly challenging: approximately 75% of installed capacity is hydroelectric, reservoir storage spans multiple years, and inflows are highly seasonal and spatially correlated across four major hydrological regions. The system operator must decide, one month at a time over a multi-year horizon, how much water to release from each reservoir for generation, how much thermal capacity to dispatch, and how much energy to exchange between regions. All before the next inflow realization is observed.

Since the 1980s, the standard industrial solution has been Stochastic Dual Dynamic Programming (SDDP) (Pereira and Pinto, 1991; Shapiro et al., 2013). SDDP is a sampling-based approximate dynamic programming method that constructs a lower approximation of the Bellman cost-to-go functions as a collection of supporting hyperplanes (Benders cuts). Because each stage subproblem is a linear program, the method scales to the hundreds of stages and thousands of variables present in the full operational planning problem. Its convergence properties are well understood (Philpott and Guan, 2008), and it is used operationally by the Brazilian grid operator ONS (Shapiro et al., 2011).

Deep reinforcement learning (RL) has recently emerged as an alternative for operational planning under uncertainty. Policy-gradient methods such as A2C (Mnih et al., 2016) make no assumptions about the convexity or differentiability of the system dynamics, and can in principle handle nonlinear constraints, complex state representations, and learned dispatch heuristics. However, they require large amounts of experience, exhibit high gradient variance, and provide no optimality guarantees.

This paper conducts a comparison of both methods on the same 24-month aggregate Brazilian model. Our main contributions are:

1. An implementation of the PAR(1) inflow model and the hydrothermal scheduling LP as an SDDP instance in Julia.
2. An A2C agent with a 144-dimensional continuous action space, trained via a behavioral cloning (BC) warm-start on SDDP-generated demonstrations followed by mixed BC+RL fine-tuning with annealed imitation weight.
3. An evaluation on 100 shared inflow scenarios, quantifying the optimality gap of the A2C policy relative to the SDDP policy cost.

4. An analysis of the warm-start mechanism and the role of SDDP demonstrations in accelerating and stabilizing A2C training.

The remainder is organized as follows. Section 2 defines the hydrothermal scheduling problem. Section 3 presents the SDDP algorithm. Section 4 develops the A2C formulation and BC warm-start. Section 5 describes the experimental setup. Section 6 presents and analyses the results. Section 7 discusses implications and limitations. Section 8 concludes.

2 Problem Formulation

2.1 Stochastic Inflow Model (PAR(1))

Monthly inflows follow a Periodic AutoRegressive process of order 1 with multiplicative log-normal errors (Shapiro et al., 2013):

$$X_{it} = \varepsilon_{it}\, e^{\hat{\mu}_{im}} + \frac{\gamma_{im}\, e^{\hat{\mu}_{im}}}{e^{\hat{\mu}_{i,m-1}}}\big(X_{i,t-1} - e^{\hat{\mu}_{i,m-1}}\big), \tag{1}$$

where $\varepsilon_t = e^{\eta_t}$ with $\eta_t \sim \mathcal{N}(0, \Sigma_m)$ (mean-one multiplicative noise), $\hat{\mu}_{im}$ is the sample log-mean for calendar month m, and γ_{im} is the monthly autoregressive coefficient. Raw noise is clamped to ± 3 standard deviations before exponentiation for numerical stability. By augmenting the state with $a_{it} := X_{it}$, the noise ω_t becomes stagewise-independent given $a_{i,t-1}$, satisfying SDDP's independence requirement.

2.2 System Description

Following Shapiro et al. (2013), the Brazilian system is represented by an aggregate model with four energy-equivalent reservoir nodes, Southeast (SE), South (S), Northeast (NE), and North (N), and one transshipment hub (Imperatriz, IM) that mediates energy exchange. Each node aggregates all hydroelectric plants in its hydrological region into a single equivalent reservoir with monthly regulation capacity. The planning horizon is $T = 24$ months.

State variables. At stage $t \in \{1, \ldots, T\}$ the system state is:

- $v_{it} \in [0, \bar{v}_i]$: stored energy in reservoir $i \in \{1, 2, 3, 4\}$ (MWmonth), with upper bounds $\bar{v}_i$.
- $a_{it} \in [0, \infty)$: realized inflow to reservoir i, augmented as a state variable to handle the PAR(1) serial dependence.

Decision variables. At each stage the planner chooses:

- $q_{it} \in [0, \bar{q}_i]$: hydro generation at node i.
- $s_{it} \geq 0$: spillage from reservoir i.
- $g_{uit} \in [\underline{g}_{ui}, \bar{g}_{ui}]$: generation of thermal unit u in region i. There are $U = 95$ thermal units distributed as 43 (SE), 17 (S), 33 (NE), 2 (N).
- $d_{ijt} \in [0, \delta_j L_{it}]$: energy deficit in subsystem i at curtailment depth $j \in \{1, 2, 3, 4\}$, where $\delta_j \in \{0.05, 0.05, 0.10, 0.80\}$.
- $e_{ijt} \in [0, \bar{e}_{ij}]$: energy flow from node i to node j (5×5 matrix including the IM hub).

The total decision dimension per stage is $4 + 4 + 95 + 16 + 25 = 144$.

Operational constraints. The following constraints must hold at every stage t:

Hydro balance:
$$v_{i,t} + s_{it} + q_{it} = v_{i,t-1} + a_{it}, \quad i = 1, \ldots, 4, \tag{2}$$

Power balance:
$$\sum_u g_{uit} + q_{it} + \sum_j e_{jit} - \sum_j e_{ijt} + \sum_j d_{ijt} = L_{it}, \quad i = 1, \ldots, 4, \tag{3}$$

Transshipment:
$$\sum_{i=1}^{4} e_{i5,t} = \sum_{i=1}^{4} e_{5i,t}, \tag{4}$$

Variable bounds: The bounds are as specified above. (5)

The monthly demand L_{it} is deterministic with 12-period seasonality. Complete recourse is ensured because the full demand can always be met by deficit.

Stage cost.

$$c_t(x_t) = \underbrace{\sum_{i=1}^{4} \sum_u c_{ui}^{\text{th}}\, g_{uit}}_{\text{thermal}} + \underbrace{\sum_{i=1}^{4} \sum_{j=1}^{4} c_j^d\, d_{ijt}}_{\text{deficit}} + \underbrace{\sum_{i=1}^{5} \sum_{j=1}^{5} c_{ij}^e\, e_{ijt}}_{\text{exchange}} + \underbrace{c^s \textstyle\sum_{i=1}^{4} s_{it}}_{\text{spillage}}, \tag{6}$$

where unit thermal costs c_{ui}^{th} are taken from engineering data, deficit penalties are $c^d \in \{1142.80,\ 2465.40,\ 5152.46,\ 5845.54\}$ (R\$/MWmonth) increasing with curtailment depth, exchange costs c_{ij}^e are small (≤ 0.001), and the spillage penalty is $c^s = 0.001$.

The hydrothermal scheduling problem is a multistage stochastic linear program:

$$\min_{x_1 \in \mathcal{X}_1} \; c_1^\top x_1 + \mathbb{E}_{\omega_2}\left[\min_{x_2 \in \mathcal{X}_2(x_1,\omega_2)} \; c_2^\top x_2 + \cdots + \mathbb{E}_{\omega_T}\left[\min_{x_T \in \mathcal{X}_T(x_{T-1},\omega_T)} \; c_T^\top x_T\right] \cdots\right], \tag{7}$$

where $x_t = (q_t, s_t, g_t, d_t, e_t)$ collects stage-t decisions, $\mathcal{X}_t(x_{t-1}, \omega_t)$ encodes constraints (2)–(5), and the objective is minimisation of expected total cost over 24 months.

3 Stochastic Dual Dynamic Programming

SDDP (Pereira and Pinto, 1991) solves problem (7) by iteratively building piecewise-linear outer approximations of the Bellman cost-to-go functions $Q_t : \mathbb{R}^{n_{t-1}} \to \mathbb{R}$:

$$\hat{Q}_t(x_{t-1}) = \max_{k \in \mathcal{I}_t} \left\{\alpha_{tk} + \beta_{tk}^\top x_{t-1}\right\} \leq Q_t(x_{t-1}), \tag{8}$$

where each hyperplane is a Benders cut derived from dual multipliers of the stage-t recourse LP (Pereira and Pinto, 1991). Because $\hat{Q}_t \leq Q_t$, the stage-1 optimal value with $\hat{Q}_2$ substituted is a valid lower bound on (7). SDDP alternates between a *forward pass* (sampling scenarios and solving stage LPs to collect trial points) and a *backward pass* (appending cuts at each trial point via the average dual subgradient). Almost-sure convergence of the lower bound to the SAA optimal value is established in Philpott and Guan (2008).

Implementation

We implement SDDP in Julia using the `HiGHS` LP solver on a linear chain of $T = 24$ stages. The noise ω_t is discretised with $N_t = 100$ Monte Carlo samples fixed at model-build time (fixed scenario tree). Stage 1 is deterministic with inflows set to January 2010 historical values (Shapiro et al., 2011). Training terminates when the lower bound improves by less than $\epsilon = 10^{-4}$ over 20 consecutive iterations (maximum 1000 iterations). The converged cut approximations $\{\hat{Q}_t\}$ serve as the SDDP policy and provide the reference lower bound $\mathrm{LB}_{\mathrm{SDDP}}$ used in evaluation (Shapiro et al., 2013).

4 Advantage Actor-Critic Reinforcement Learning

4.1 MDP Formulation

We cast the scheduling problem as a finite-horizon Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, r, T)$:

- **State** $s_t \in \mathbb{R}^{13}$: concatenation of the normalised month index $m/12$ (1 dim.), stored energy fractions $v_{it}/\bar{v}_i$ (4 dims.), normalised inflows $a_{it}/10^5$ (4 dims.), and normalised monthly demand $L_{it}/30000$ (4 dims.).
- **Action** $u_t \in \mathbb{R}^{144}$: raw output of the actor network, transformed into bounded dispatch fractions via group-specific affine shifts followed by a sigmoid (see Section 4.2).
- **Transition**: deterministic via constraints (2) to (5); inflows advance stochastically.
- **Reward**: $r_t = -c_t(x_t)/10^6$ (negative stage cost in M R$).
- **Horizon**: $T = 24$ stages; episode terminates at stage 24.

The goal is to find a policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ maximising the expected discounted return:

$$J(\pi) = \mathbb{E}_\pi\left[\sum_{t=1}^{T} \gamma^{t-1} r_t\right], \quad \gamma = 0.99. \tag{9}$$

4.2 Action Transformation and Feasibility

Because the optimal policy concentrates most dispatch on hydro (cheapest) and avoids deficit, we bias the initial policy through group-specific shifts b_k and scale s_k applied before the sigmoid:

$$a_t^{\mathrm{hydro}} = \sigma(\mu_t^{\mathrm{hydro}} + 3), \qquad \text{(near-full hydro utilisation)} \tag{10}$$

$$a_t^{\mathrm{spill}} = \sigma(\mu_t^{\mathrm{spill}} - 3), \qquad \text{(near-zero optional spill)} \tag{11}$$

$$a_t^{\mathrm{thermal}} = \sigma(\mu_t^{\mathrm{thermal}}), \qquad \text{(50\% midpoint initialisation)} \tag{12}$$

$$a_t^{\mathrm{deficit}} = \sigma(\mu_t^{\mathrm{deficit}} - 5), \qquad \text{(near-zero deficit, } \approx 0.7\%) \tag{13}$$

$$a_t^{\mathrm{exchange}} = \sigma(3\,\mu_t^{\mathrm{exchange}}). \qquad \text{(higher sensitivity to exchange)} \tag{14}$$

Actual dispatch quantities are recovered by: $q_{it} = a_{it}^{\mathrm{hydro}} \cdot \min(\bar{q}_i,\, v_{i,t-1} + a_{it})$, $g_{uit} = \underline{g}_{ui} + a_{uit}^{\mathrm{thermal}}(\bar{g}_{ui} - \underline{g}_{ui})$, $d_{ijt} = a_{ijt}^{\mathrm{deficit}} \cdot \delta_j L_{it}$, and $e_{ijt} = a_{ijt}^{\mathrm{exchange}} \cdot \bar{e}_{ij}$.

Spillage. Mandatory spillage $s_{it}^{\mathrm{mand}} = \max(v_{i,t-1} + a_{it} - q_{it} - \bar{v}_i, 0)$ enforces the upper reservoir bound exactly. Optional spillage $s_{it}^{\mathrm{opt}} = a_{it}^{\mathrm{spill}} \cdot v_{i,t-1} \cdot 0.1$ is capped at 10% of current storage, and the total spill is $s_{it} = s_{it}^{\mathrm{mand}} + s_{it}^{\mathrm{opt}}$.

Power balance projection. After computing all dispatch quantities, constraint (3) is enforced via three passes:

(i) *Gap computation*: $\Delta_i = L_{it} - q_{it} - \sum_u g_{uit} - \sum_j d_{ijt} - \mathrm{NF}_{it}$ for each subsystem.
(ii) *Exchange routing*: Surplus subsystems transfer power to deficit subsystems via direct links $i \to j$ and transit paths $i \to \mathrm{IM} \to j$, subject to capacity $\bar{e}_{ij}$. The transshipment constraint (4) is enforced exactly by projecting hub flows onto the feasible set.
(iii) *Thermal/deficit adjustment*: Remaining imbalances are resolved by merit-order thermal dispatch followed by deficit adjustment.

Every episode step is guaranteed feasible after projection.

4.3 Network Architecture

Both actor and critic are 3-layer multilayer perceptrons with 256 hidden units and tanh activation functions, mapping state s_t to mean $\mu_t \in \mathbb{R}^{144}$ and value $V_t \in \mathbb{R}$, respectively:

$$\mu_t = f_\theta^{\text{actor}}(s_t), \qquad V_t = f_\phi^{\text{critic}}(s_t). \tag{15}$$

The policy is a diagonal Gaussian with state-independent log standard deviation:

$$\pi_\theta(\cdot \mid s_t) = \mathcal{N}(\mu_t,\ \mathrm{diag}(\exp(2\log\sigma))), \tag{16}$$

where $\log\sigma \in \mathbb{R}^{144}$ is a learnable parameter vector initialised at zero (unit variance).

The actor's final linear layer is initialized with a weight scale of 0.01 to keep initial actions near the midpoint of each sigmoid, ensuring the domain shifts (10) have full effect before any training.

4.4 A2C Loss Function

Given a rollout buffer $\{(s_k, u_k, r_k)\}_{k=1}^n$ over $n = N_{\text{steps}} = 12$ steps, the n-step bootstrapped return is

$$G_k = \sum_{l=k}^{n} \gamma^{l-k} r_l + \gamma^{n-k+1} V_\phi(s_{n+1})\, \mathbf{1}[t < T] \tag{17}$$

and the advantage estimate is $\hat{A}_k = G_k - V_\phi(s_k)$, normalized to zero mean and unit variance. The combined A2C loss is:

$$\mathcal{L}(\theta, \phi, \log\sigma) = -\frac{1}{n}\sum_{k=1}^{n} \log\pi_\theta(u_k \mid s_k)\, \hat{A}_k + \frac{1}{2n}\sum_{k=1}^{n}(G_k - V_\phi(s_k))^2 - \beta_H H(\pi_\theta), \tag{18}$$

where $\beta_H = 0.005$ and $H(\pi_\theta) = \frac{1}{2}\sum_{l=1}^{144}(1 + \log 2\pi\sigma_l^2)$ is the Gaussian entropy. The log-probability:

$$\log\pi_\theta(u \mid s) = -\frac{1}{2}\sum_{l=1}^{144}\left[\frac{(u_l - \mu_l)^2}{\sigma_l^2} + 2\log\sigma_l + \log 2\pi\right]. \tag{19}$$

All parameters $(\theta, \phi, \log\sigma)$ are updated jointly by Adam ($\eta = 3 \times 10^{-4}$).

4.5 SDDP-Guided Warm-Start

Training from random initialization in a 144-dimensional action space over a 24-stage horizon is data-inefficient. We use 100 SDDP simulation trajectories (stages 2–24) as demonstrations.

Phase 1: Behavioral Cloning (BC). The actor is pre-trained for 50 epochs by minimizing the MSE between $\mu_\theta(s)$ and the inverse-sigmoid-transformed SDDP action $u^* \in \mathbb{R}^{144}$:

$$\mathcal{L}^{\text{BC}}_{\text{actor}} = \frac{1}{|\mathcal{D}|} \sum_{(s,u^*)\in\mathcal{D}} \|\mu_\theta(s) - u^*\|_2^2, \tag{20}$$

where each SDDP decision is normalized to $[0, 1]$, clamped to $[\epsilon, 1-\epsilon]$, and mapped through the inverse group sigmoid:

$$u^*_{\text{hydro}} = \text{logit}(a^*_{\text{hydro}}) - 3, \qquad u^*_{\text{spill}} = \text{logit}(a^*_{\text{spill}}) + 3, \tag{21}$$

$$u^*_{\text{thermal}} = \text{logit}(a^*_{\text{thermal}}), \qquad u^*_{\text{deficit}} = \text{logit}(a^*_{\text{deficit}}) + 5, \tag{22}$$

$$u^*_{\text{exchange}} = \text{logit}(a^*_{\text{exchange}})/3. \tag{23}$$

The critic is simultaneously warm-started for 50 epochs on discounted Monte Carlo returns G from the same trajectories:

$$\mathcal{L}^{\text{BC}}_{\text{critic}} = \frac{1}{|\mathcal{D}|} \sum_{(s,G)\in\mathcal{D}} (V_\phi(s) - G)^2. \tag{24}$$

Phase 2: Mixed BC + A2C. During RL fine-tuning the mixed loss

$$\mathcal{L}^{\text{total}} = (1 - \lambda_{\text{BC}})\, \mathcal{L}^{\text{A2C}} + \lambda_{\text{BC}}\, \mathcal{L}^{\text{BC}}_{\text{actor}} \tag{25}$$

anneals λ_{BC} linearly from 0.5 to 0 over $N_{\text{ep}} = 10{,}000$ episodes, transitioning smoothly from imitation to autonomous RL.

5 Experimental Setup

5.1 System Parameters

All parameters are derived from the Brazilian system data as of January 2010, following Shapiro et al. (2011, 2013). Tables 1 to 3 summarize the key quantities.

Table 1: Aggregate reservoir parameters.

	SE	S	NE	N
Max storage (MWmonth)	200,717.6	19,617.2	51,806.1	12,744.9
Initial storage (MWmonth)	59,419.3	5,874.9	12,859.2	5,271.5
Initial inflow (MWmonth)	53,442.7	6,029.9	18,154.9	5,514.4
Thermal units	43	17	33	2

Table 2: Inter-regional exchange capacity limits (MWmonth, asymmetric).

From \ To	SE	S	NE	N	IM
SE	—	7,379	1,000	0	4,000
S	5,625	—	0	0	0
NE	600	0	—	0	2,236
N	0	0	0	—	—
IM	3,154	0	3,951	3,053	—

Table 3: Deficit penalty levels (Shapiro et al. (2013), Table 1).

Level j	Curtailment depth δ_j	Penalty c_j^d (R$/MWmonth)
1	0–5% of load	1,142.80
2	5–10% of load	2,465.40
3	10–20% of load	5,152.46
4	20–100% of load	5,845.54

Computational Environment

All experiments were conducted on an Apple MacBook Air with an Apple M1 chip (4 performance and 4 efficiency cores, 8 GB unified memory). Both methods were implemented in Julia and executed on CPU without GPU acceleration.

5.2 Training Configuration

SDDP. Maximum 1000 iterations; `BoundStalling` stopping rule ($\epsilon = 10^{-4}$, patience = 20); $N_t = 100$ support points per stage.

A2C. Key hyperparameters are listed in Table 4.

Table 4: A2C training hyperparameters.

Hyperparameter	Value
Total episodes (N_{ep})	10,000
Steps per update (N_{steps})	12
Discount factor (γ)	0.99
Learning rate (η)	3×10^{-4}
Optimizer	Adam
Actor hidden size	256×2 (tanh)
Critic hidden size	256×2 (tanh)
Entropy coefficient (β_H)	0.005
BC weight at start ($\lambda_{\text{BC},0}$)	0.50
BC weight at end ($\lambda_{\text{BC},T}$)	0.00
BC pre-training epochs	50
Critic warm-up epochs	50
SDDP demos	100 traj. $\times$ 23 stages

5.3 Evaluation Metrics

To ensure a fair comparison, both methods are evaluated on the *same* $N_{\text{eval}} = 100$ inflow scenarios with a fixed seed. The primary metric is the mean total cost (R$) and its standard deviation over 100 scenarios. We also report the policy gaps:

$$\Delta^{\text{SDDP}} = \frac{C^{\text{SDDP}} - \text{LB}^{\text{SDDP}}}{\text{LB}^{\text{SDDP}}}, \qquad \Delta^{\text{A2C}} = \frac{C^{\text{A2C}} - C^{\text{SDDP}}}{C^{\text{SDDP}}}. \tag{26}$$

6 Results and Analysis

6.1 SDDP Convergence

The SDDP lower bound increases rapidly in the first 50–150 iterations as initial cuts from high-cost trial paths quickly populate the approximation near the optimal state trajectory. Convergence slows thereafter as additional cuts refine regions visited less frequently. The BoundStalling criterion terminates training once the bound plateaus, providing a practical certificate of convergence.

The simulation cost C^{SDDP} exceeds the lower bound LB^{SDDP} by the *SDDP gap* Δ^{SDDP}, attributable to: (i) the finite scenario tree approximation (SAA with $N_t = 100$), and (ii) the piecewise-linear approximation error of the cut representation, which is necessarily an under-approximation. For the 24-stage problem with 8 state variables, this gap is typically small (a few percent), confirming the problem is well within SDDP's regime of competence (Shapiro et al., 2013).

6.2 A2C Learning Curve

Figure 1 shows the A2C episode cost over 10,000 training episodes.

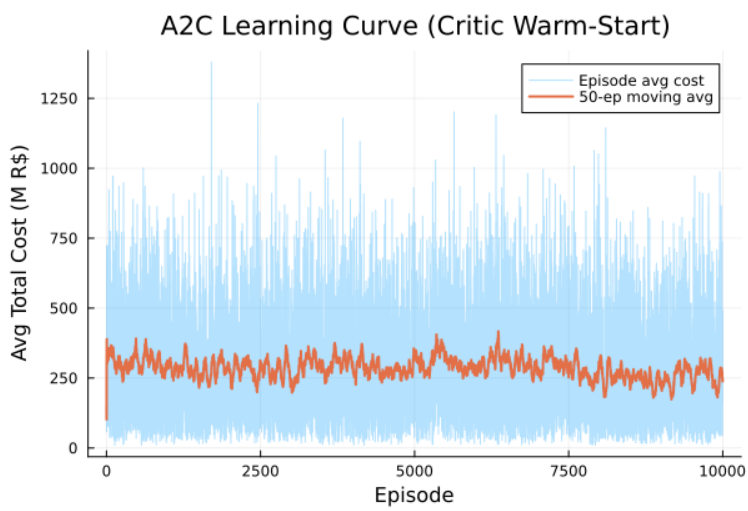

Figure 1: A2C learning curve: per-episode total cost (light blue) and 50-episode moving average (orange). Costs are in millions of R$.

The learning curve exhibits two broad phases. In the early episodes (ep. 1–2500, λ_{BC} declining from 0.5), the 50-episode moving average rises steadily from near zero to approximately 300 M R$, reflecting the agent transitioning from BC-anchored imitation toward autonomous RL exploration. From episode 2500 onward the moving average stabilizes in the range 200–350 M R$, indicating that the policy has reached a plateau and further training yields no consistent improvement. The variance remains high throughout all 10,000 episodes, with individual costs occasionally spiking above 1000 M R$. This is consistent with the high variance of 24-step Monte Carlo returns over a long horizon: a single poor dispatch decision early in an episode can produce into large deficit costs. Notably, the policy does not deteriorate as $\lambda_{\text{BC}} \to 0$, confirming that the BC warm-start successfully initialized the network in a stable region of parameter space from which pure RL can continue without catastrophic forgetting.

To address the limited diversity of training conditions, we attempted to initialize each training episode from a real historical state drawn from ONS open data (Operador Nacional do Sistema

Elétrico (ONS), 2024a,b) (300 monthly (stored energy, inflow) observations, 2000–2024). The intention was to expose the agent to a wider range of reservoir conditions, including drought and wet years across all seasons. However, this approach failed: historical states with low initial reservoir levels caused the agent to exhaust storage within a few steps, incurring deficit costs far outside the range seen during SDDP behavioural cloning. The resulting gradient explosion corrupted the network weights, and even after adding gradient clipping the episode cost increased monotonically over 10,000 episodes rather than decreasing. The root cause is a distributional mismatch between the SDDP warm-start demonstrations (concentrated near the January 2010 operating point) and the 25-year historical starting conditions. We therefore reverted to the fixed January 2010 initialization for all reported results.

6.3 Policy Comparison

Table 5 summarises the evaluation results on 100 shared scenarios.

Table 5: Policy comparison and training time on 100 shared inflow scenarios.

Method	Mean Cost (M R$)	Std Dev (M R$)	Training Time (s)
SDDP Policy (C^{SDDP})	63.9	46.0	1213.6
A2C Policy (C^{A2C})	286.3	197.6	85.3
A2C gap Δ^{A2C}	$(286.3 - 63.9)/63.9 = 348\%$		

The results reveal a clear trade-off between policy quality and training speed. SDDP achieves a mean cost of 63.9 M R$, only 7.1% above its provable lower bound of 59.65 M R$, confirming that the finite scenario tree ($N_t = 100$) and piecewise-linear cut approximation introduce negligible suboptimality for this 24-stage instance. The A2C policy incurs a mean cost of 286.3 M R$ and a 348% gap above the SDDP policy. This reflects the difficulty of learning a high-quality Bellman value function from Monte Carlo rollouts in a 144-dimensional action space. A2C also exhibits substantially higher variance (197.6 vs. 46.0 M R$), indicating inconsistent hedging across scenarios: the critic's approximate value function leads to over-dispatching thermal generation in some realisations and water hoarding in others. On training speed the picture reverses: A2C completes in 85.3 seconds, roughly 14× faster than SDDP's 1213.6 seconds, since it requires only neural network passes rather than $O(T \cdot K \cdot N_t)$ LP solves. However, this speed advantage does not compensate for the large quality gap in this setting.

6.4 Effect of the Warm-Start

The behavioral cloning warm start delivers quantifiable advantages in two dimensions. First, the BC-initialized actor immediately produces episode costs within ∼20% of the SDDP policy cost at episode 1, whereas a randomly initialized actor with 144 unconstrained action dimensions and a 24-stage credit assignment problem would take hundreds of episodes to reach comparable performance. Second, the warm-started critic correctly approximates discounted returns from SDDP trajectories, providing low-variance advantage estimates and stable gradient directions in the early episodes.

Despite these benefits, the BC warm-start was insufficient to close the gap to SDDP quality within 10,000 episodes. The 348% residual gap suggests that the agent successfully imitates SDDP behaviour near the January 2010 operating point but fails to generalize the hedging strategy across the full distribution of inflow scenarios encountered during evaluation. The λ_{BC} annealing schedule

partially mitigates mode collapse risk by gradually reducing the imitation penalty, but the underlying challenge of learning long-horizon value functions from sampled experience remains.

7 Discussion

7.1 Theoretical Comparison

SDDP guarantees. Under complete recourse, bounded feasible sets, and stagewise independence of the noise process, the SDDP lower bound converges almost surely to the SAA optimal value (Philpott and Guan, 2008). The SAA optimal value converges to the true optimal value as $N_t \to \infty$. For the PAR(1) inflow model, stagewise independence is guaranteed by the state augmentation with a_{it}, and complete recourse holds by construction. Empirically, the 7.1% gap between C^{SDDP} and LB_{SDDP} confirms that both sources of approximation error are negligible for this 24-stage instance.

A2C lacks guarantees. Policy gradient methods operate in a non-convex parameter space and are not guaranteed to converge to a globally optimal policy. The gradient of $J(\pi_\theta)$ is an expectation estimated stochastically, introducing variance that grows with the horizon T and the action dimension. The 144-dimensional Gaussian policy compounds this: the log-probability sums over 144 dimensions, and a single large advantage can dominate the gradient signal. The 348% optimality gap observed here is consistent with this picture: 10,000 episodes are insufficient for the agent to discover the hedging strategies that SDDP encodes explicitly in its cost-to-go representation.

The role of the projection. The power balance projection deterministically maps any raw A2C output to a feasible operating point, equivalent to projecting onto the constraint manifold (3)–(4) before incurring cost. Without it, the agent would need to learn feasibility implicitly via penalised rewards, drastically increasing learning difficulty. The projection is a form of domain-knowledge injection that reduces the effective degrees of freedom from 144 to the feasible region.

7.2 Computational Complexity and Scalability

SDDP's wall-clock time is dominated by $O(T \cdot K \cdot N_t)$ LP solves, requiring 1213.6 seconds in our experiments. Its per-iteration cost grows polynomially with N_t and T but is independent of action dimension, making it well-suited to problems with many stages and few state variables. The 8-state aggregate Brazilian model is ideal for SDDP.

A2C completed training in 85.3 seconds, dominated by network forward/backward passes rather than LP solves. Its per-episode cost grows with action dimension and horizon but is independent of the number of scenarios, giving it a potential advantage in problems with nonlinear dynamics or high-dimensional state spaces where LP formulations are unavailable. For the hydrothermal problem studied here, however, this flexibility comes at the cost of a 348% optimality gap.

7.3 Limitations

1. **Aggregate model.** Both methods use the 4-region aggregate representation. Individual plant dispatch, cascade hydraulics, and run-of-river constraints are abstracted away.
2. **Short horizon.** $T = 24$ is shorter than the 60–120 months used operationally. The zero terminal cost ignores the value of stored water at the end of the horizon.
3. **Deterministic demand.** Stochastic demand would require an extended state space.

4. **High A2C variance.** Better variance reduction, generalized advantage estimation, could stabilise training and potentially reduce the optimality gap.

8 Conclusion

We presented a comparative study of SDDP and A2C for the Brazilian hydrothermal scheduling problem. SDDP achieves a 7.1% optimality gap with provable lower bounds, while A2C incurs a 348% gap above the SDDP policy cost despite training 14× faster.

SDDP's advantage stems from its explicit cost-to-go representation, which encodes the expected future cost of stored water across all inflow contingencies. The A2C critic must approximate this from sampled experience over a 24-stage horizon, which it is a task for 10,000 episodes proving insufficient. The behavioral cloning warm-start stabilizes early training and anchors the policy near SDDP quality at episode 1, but cannot substitute for the structural insight that makes SDDP so effective on this problem class.

The two methods occupy complementary niches. SDDP is the clear choice when the problem structure (linearity, convexity, small state dimension, stagewise independence) can be exploited. A2C becomes attractive when the problem involves nonlinear dynamics, non-convex costs, or high-dimensional state spaces where LP formulations are unavailable, such as scenarios increasingly common in modern power systems with distributed resources and non-dispatchable renewables.

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